

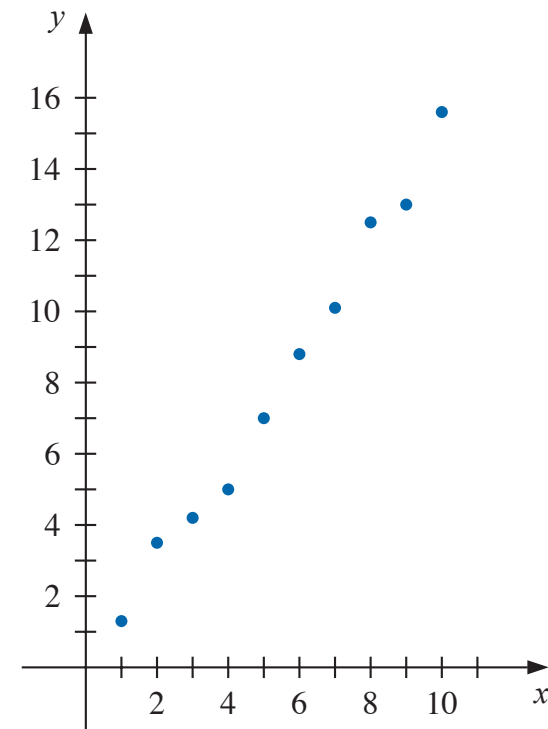
Section 5

Approximation Theory

Least squares approximation

Given N data points $\{(x_i, y_i)\}$ for $i = 1, \dots, N$, can we determine a linear model $y = a_1x + a_0$ (i.e., find a_0, a_1) that fits the data?

x_i	y_i	x_i	y_i
1	1.3	6	8.8
2	3.5	7	10.1
3	4.2	8	12.5
4	5.0	9	13.0
5	7.0	10	15.6



Matrix formulation

We can simplify notations by using matrices and vectors:

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^N, \quad X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_N \end{bmatrix} \in \mathbb{R}^{N \times 2}$$

So we want to find $a = (a_0, a_1)^\top \in \mathbb{R}^2$ such that $y \approx Xa$.

Several types of fitting criteria

There are several types of criteria for “best fitting”:

- ▶ Define the error function as

$$E_{\infty}(a) = \|y - Xa\|_{\infty}$$

and find $a^* = \arg \min_a E_{\infty}(a)$. This is also called the **minimax** problem since the problem $\min_a E_{\infty}(a)$ can be written as

$$\min_a \max_{1 \leq i \leq n} |y_i - (a_0 + a_1 x_i)|$$

- ▶ Define the error function as

$$E_1(a) = \|y - Xa\|_1$$

and find $a^* = \arg \min_a E_1(a)$. E_1 is also called the **absolute deviation**.

Least squares fitting

In this course, we focus on the widely used **least squares**.

Define the least squares error function as

$$E_2(a) = \|y - Xa\|_2 = \sum_{i=1}^n |y_i - (a_0 + a_1 x_i)|^2$$

and the least squares solution a^* is

$$a^* = \arg \min_a E_2(a)$$

Least squares fitting

To find the optimal parameter a , we need to solve

$$\nabla E_2(a) = 2X^\top(Xa - y) = 0$$

This is equivalent to the so-called **normal equation**:

$$X^\top Xa = X^\top y$$

Note that $X^\top X \in \mathbb{R}^{2 \times 2}$ and $X^\top y \in \mathbb{R}^2$, so the normal equation is easy to solve!

Least squares fitting

It is easy to show that

$$X^{\top} X = \begin{bmatrix} N & \sum_{i=1}^N x_i \\ \sum_{i=1}^N x_i & \sum_{i=1}^N x_i^2 \end{bmatrix}, \quad X^{\top} y = \begin{bmatrix} \sum_{i=1}^N y_i \\ \sum_{i=1}^N x_i y_i \end{bmatrix}$$

Using the close-form of inverse of 2-by-2 matrix, we have

$$(X^{\top} X)^{-1} = \frac{1}{N \sum_{i=1}^N x_i^2 - (\sum_{i=1}^N x_i)^2} \begin{bmatrix} \sum_{i=1}^N x_i^2 & -\sum_{i=1}^N x_i \\ -\sum_{i=1}^N x_i & N \end{bmatrix}$$

Least squares fitting

Therefore we have the solution

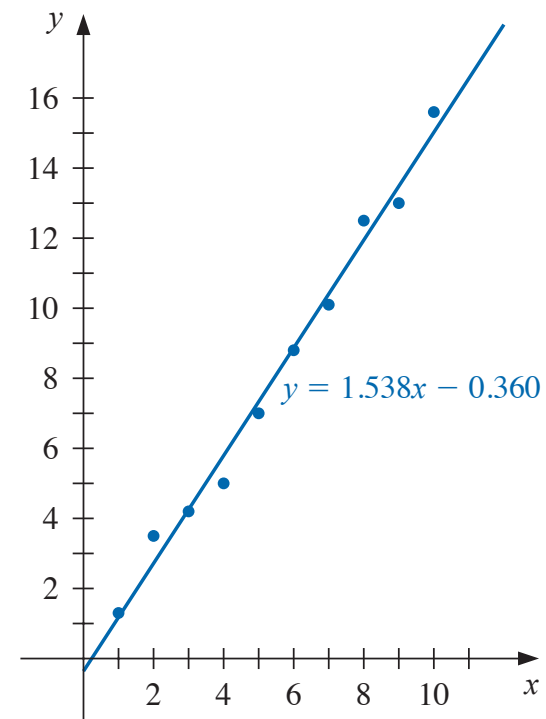
$$\begin{aligned} a^* &= \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = (X^\top X)^{-1} (X^\top y) \\ &= \begin{bmatrix} \frac{\sum_{i=1}^N x_i^2 \sum_{i=1}^N y_i - \sum_{i=1}^N x_i y_i \sum_{i=1}^N x_i}{N \sum_{i=1}^N x_i^2 - (\sum_{i=1}^N x_i)^2} \\ \frac{N \sum_{i=1}^N x_i y_i - \sum_{i=1}^N x_i \sum_{i=1}^N y_i}{N \sum_{i=1}^N x_i^2 - (\sum_{i=1}^N x_i)^2} \end{bmatrix} \end{aligned}$$

Least squares fitting

Example

Least squares fitting of the data gives $a_0 = -0.36$ and $a_1 = 1.538$.

x_i	y_i	x_i	y_i
1	1.3	6	8.8
2	3.5	7	10.1
3	4.2	8	12.5
4	5.0	9	13.0
5	7.0	10	15.6



Polynomial least squares

The least squares fitting presented above is also called **linear least squares** due to the linear model $y = a_0 + a_1x$.

For general least squares fitting problems with data $\{(x_i, y_i) : i = 1, \dots, N\}$, we may use polynomial

$$P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

as the fitting model. Note that $n = 1$ reduces to linear model.

Now the **polynomial least squares** error is defined by

$$E(a) = \sum_{i=1}^N |y_i - P_n(x_i)|^2$$

where $a = (a_0, a_1, \dots, a_n)^\top \in \mathbb{R}^{n+1}$.

Matrices in polynomial least squares fitting

Like before, we use matrices and vectors:

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^N, \quad X = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^n \\ 1 & x_2 & x_2^2 & \cdots & x_2^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_N & x_N^2 & \cdots & x_N^n \end{bmatrix} \in \mathbb{R}^{N \times (n+1)}$$

So we want to find $a = (a_0, a_1, \dots, a_n)^\top \in \mathbb{R}^{n+1}$ such that $y \approx Xa$.

Polynomial least squares fitting

Same as above, we need to find a such that

$$\nabla E_2(a) = 2X^\top(Xa - y) = 0$$

which has **normal equation**:

$$X^\top Xa = X^\top y$$

Note that now $X^\top X \in \mathbb{R}^{(n+1) \times (n+1)}$ and $X^\top y \in \mathbb{R}^{n+1}$. From normal equation we can solve for the fitting parameter

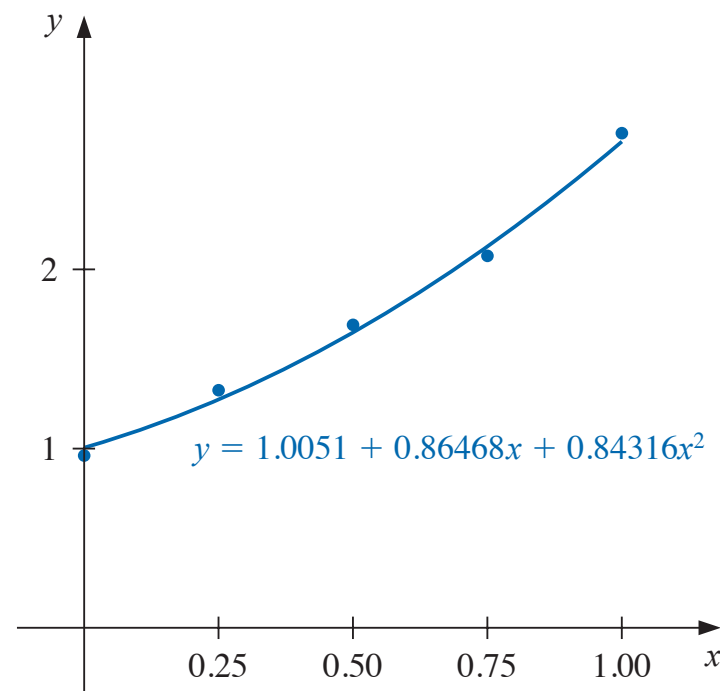
$$a^* = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = (X^\top X)^{-1}(X^\top y)$$

Polynomial least squares

Example

Least squares fitting of the data using $n = 2$ gives $a_0 = 1.0051$, $a_1 = 0.86468$, $a_2 = 0.84316$.

i	x_i	y_i
1	0	1.0000
2	0.25	1.2840
3	0.50	1.6487
4	0.75	2.1170
5	1.00	2.7183



Other least squares fitting models

In some situations, one may design model as

$$y = be^{ax}$$

$$y = bx^a$$

as well as many others.

To use least squares fitting, we note that they are equivalent to, respectively,

$$\log y = \log b + ax$$

$$\log y = \log b + a \log x$$

Therefore, we can first convert (x_i, y_i) to $(x_i, \log y_i)$ and $(\log x_i, \log y_i)$, and then apply standard linear least squares fitting.

Approximating functions

We now consider fitting (approximation) of a given function

$$f(x) \in C[a, b]$$

Suppose we use a polynomial $P_n(x)$ of degree n to fit $f(x)$, where

$$P_n(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

with fitting parameters $a = (a_0, a_1, \dots, a_n)^\top \in \mathbb{R}^{n+1}$. Then the least squares error is

$$E(a) = \int_a^b |f(x) - P_n(x)|^2 dx = \int_a^b \left| f(x) - \sum_{k=0}^n a_k x^k \right|^2 dx$$

Approximating functions

The fitting parameter a needs to be solved from $\nabla E(a) = 0$.

To this end, we first rewrite $E(a)$ as

$$E(a) = \int_a^b (f(x))^2 dx - 2 \sum_{k=0}^n a_k \int_a^b x^k f(x) dx + \int_a^b \left(\sum_{k=0}^n a_k x^k \right)^2 dx$$

Therefore $\nabla E(a) = \left(\frac{\partial E}{\partial a_0}, \frac{\partial E}{\partial a_1}, \dots, \frac{\partial E}{\partial a_n} \right)^\top \in \mathbb{R}^{n+1}$ where

$$\frac{\partial E}{\partial a_j} = -2 \int_a^b x^j f(x) dx + 2 \sum_{k=0}^n a_k \int_a^b x^{j+k} dx$$

for $j = 0, 1, \dots, n$.

Approximating functions

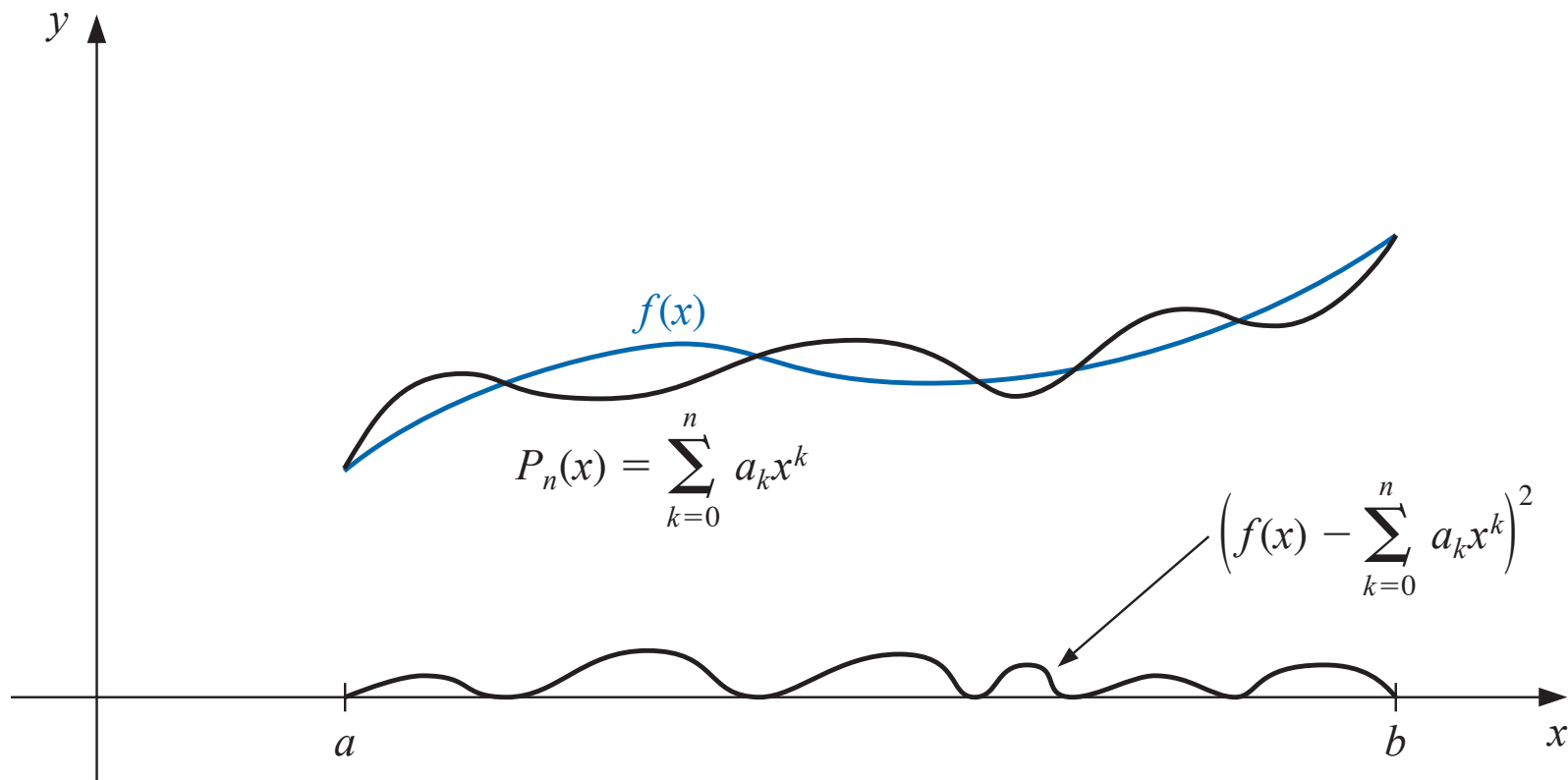
By setting $\frac{\partial E}{\partial a_j} = 0$ for all j , we obtain the **normal equation**

$$\sum_{k=0}^n \left(\int_a^b x^{j+k} dx \right) a_k = \int_a^b x^j f(x) dx$$

for $j = 0, \dots, n$. This is a linear system of $n + 1$ equations, from which we can solve for $a^* = (a_0, \dots, a_n)^\top$.

Approximating functions

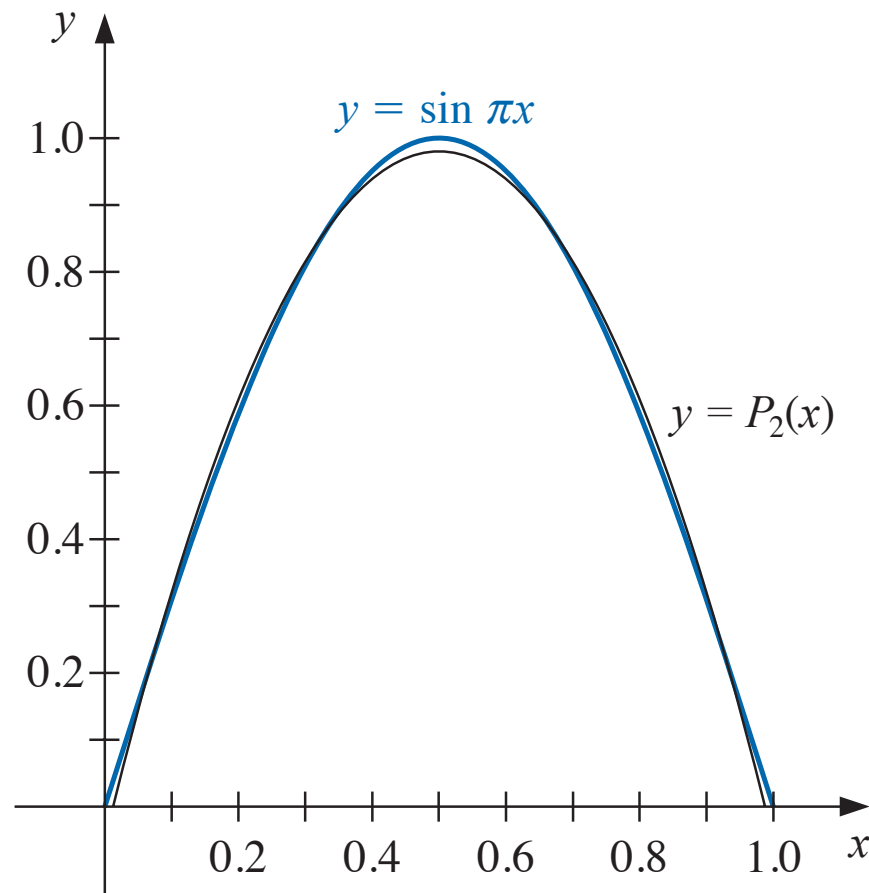
For the given function $f(x) \in C[a, b]$, we obtain least squares approximating polynomial $P_n(x)$:



Approximating functions

Example

Use least squares approximating polynomial of degree 2 for the function $f(x) = \sin(\pi x)$ on the interval $[0, 1]$.



Least squares approximations with polynomials

Remark

- ▶ The matrix in the normal equation is called **Hilbert matrix**, with entries of form

$$\int_a^b x^{j+k} dx = \frac{b^{j+k+1} - a^{j+k+1}}{j+k+1}$$

which is prone to round-off errors.

- ▶ The parameters $a = (a_0, \dots, a_n)^\top$ we obtained for polynomial $P_n(x)$ cannot be used for $P_{n+1}(x)$ – we need to start the computations from beginning.